

A Online Appendix

A.1 Detailed Description of Operations in The Central Mediterranean

In this section we provide a detailed overview of Search and Rescue Operations from the beginning of the high refugee inflow that started in 2013.

Operation Hermes Operation Hermes was a joint sea operation carried out by the armed forces of several member states of the EU and coordinated by Frontex, the European Border and Coast Guard Agency. The operation has been in place since 2008 and has since been renewed with slight modifications until October 2013. Operation Hermes had the objective to conduct border surveillance and monitor illegal crossings in the Mediterranean. It was financed with less than 1 million euros per month, and its operational area did only extend for roughly 45 km after the Sicilian coasts, in the South of Italy (European Commission 2013).

Operation Mare Nostrum Mare Nostrum was an operation carried out by the Italian Government. The operation started after the shipwreck of October 2013 that killed more than 360 migrants and has been the only one with the explicit goal of safeguarding human life at sea. It was financed with 9.5 million euros per month, and its operational area extended virtually to Libyan territorial waters. Mare Nostrum has been the most controversial operation from a political standpoint, being praised for its humanitarian impact and criticized for creating an unintended *pull factor* for migrants (Deiana et al. in press). The high costs and controversial aspects of operation Mare Nostrum led the Italian government to ask for a joint effort of the EU in making the Mediterranean a safer route.

Operation Triton Operation Triton was an EU-led operation that tried to bring together the opposing views about Mare Nostrum (ANSA 2014). The operation had the official goal of patrolling the external borders of the EU. However, it had to follow the obligations of international and European law, which required intervention in case of sightings of boats in

distress. The operation started in November 2014 and went through two phases: a first phase during which the operational area only extended for 56 km from Italian and Maltese coasts, and a second phase, starting in April 2015, which saw a large expansion of the operational area to 256 km from the Italian coasts. Operation Triton ended in February 2018 and was substituted by operation Themis, which is still running and is similar to operation Triton but with a major focus on law enforcement instead of search and rescue activity.

A.2 Additional Data

Arrivals in Europe and Departures from Libya To analyze how migration flows to Europe evolved over time, we use UNHCR data on migrants’ arrivals in Italy and Spain. This data comes from national Coast Guard authorities. We observe the monthly numbers of arrivals both by sea and land. For each month, we also observe the migrants’ country of origin. We use this data to analyze if there was a substitution between the Central Mediterranean Route – with migrants arriving to Italy – and the Western Mediterranean Route – with migrants arriving to Spain –.

Arrivals to Italy reached their peak in 2014-2016 at 150.000 migrants a year. Migrant deaths and disappearances reached their peak during the same years, with almost 2.000 deaths per year. During 2017 – the year when the MoU was signed– arrivals fell by roughly 90%. On the other hand, arrivals in Spain started sharply increasing after 2017, going from roughly 2000 per month before the signing of the MoU to almost 5000 during November 2017.

We complement data on migrant arrivals with data on estimated departures from Libya. The estimated number of departures from Libya comes from the Italian Institute for International Political Studies (ISPI).

Migration flows in North Africa To capture the presence of migrants on the routes to the Mediterranean, we use the IOM’s Flow Monitoring data for Niger. This data tracks the movements of migrants at key transit points over time. We use this data to identify the magnitude and direction of the most relevant migration flows. In some cases, the information also includes surveys of the main reason for migrating. In particular, we focus on data from the transit points in Niger – a key transit country for reaching Libya from other African countries –, where migrants are asked if they intend to continue their journey to Europe through Libya or Algeria

NGOs’ search and rescue operations After 2015 search and rescue activities by NGOs in the Mediterranean strongly intensified. We use data collected by the IOM (Cusumano & Villa, 2020), which contains a list of active NGOs along with the number of boats operating

in the Mediterranean at a monthly frequency as well as information about the ship size. We use this data to control for the presence of NGOs in the Central Mediterranean.

As Figure A.8 in the Appendix, shows, NGOs' activities keep growing from the end of 2014 to the start of 2017. The first NGO observed started its operations around September 2014. Between 2017 and the end of 2018, the number of active ships stabilizes between 15 and 18.

Maritime traffic We use monthly data from the Suez Canal Authority to proxy for the number of commercial ships sailing in the Mediterranean during our period of analysis. The Suez Canal provides the fastest crossing between the Atlantic Ocean and Indian Ocean, and is extensively used by modern commercial ships.¹¹ These data provide us with the number of ships passing through the Suez Canal as well as their directions — northbound or southbound — which allows us to estimate how many vessels are crossing the Mediterranean Sea in any given month. We use this data to ensure that our effects are not driven by commercial vessels changing their routes because of the signing of the MoU.

Sea conditions We retrieve data on sea conditions from the European Centre for Medium-Range Weather Forecast (ECMWF). We follow Deiana et al. (in press) and Battiston (2022), and use data on the significant height of combined wind waves and swell as a proxy for the roughness of sea conditions within a specific month and quadrant. This data will allow us to control for the roughness of the sea, in order to ensure that our estimates are not driven by negative weather shocks that are contemporaneous with the signing of the MoU.

Conflict data We use information about the conflict in Libya from the Uppsala Conflict Data Program (UCDP). The UCDP is one of the world's most comprehensive providers of data on organized violence and allows us to reconstruct a time series of all violent events which led to at least one death in Libya since the start of the First Civil War. The main sources of the information are media articles, NGO and international organizations' reports, case studies, official commissions' analyses, and historical archives. We use this data to

¹¹<https://www.suezcanal.gov.eg/English/About/SuezCanal/Pages/AboutSuezCanal.aspx>

ensure that our estimates are not biased by conflict-related events that might coincide with the signing of the MoU

Data on Type of Vessel We obtain data on the type of vessels that migrants use to cross the Mediterranean. We use this data to control for the fact that smugglers can anticipate the externalization of SAR activities to the LCG and switch to unsafer boats before tighter border control is enforced. In particular we are interested in the monthly share of rubber boats – which are less safe than sturdier boats — over the total number of boats. The data is obtained from [Deiana et al. \(2024\)](#)

A.3 Data Limitations

Our main outcome variable – the death incidents occurred while migrating – comes from the missing migrants project by the IOM. While this data is considered to be the largest and most comprehensive source on deaths at sea it comes with its own limitations.

We do not observe unauthorized flows, as they are by definition, undocumented. What we observe is the number of people who have been found dead. The number of people found dead is a function of three variables: the number of departures, the probability of being found, and the probability of dying. We can not observe any of these three variables. Crucially, however, as long as the variables determining the number of people found dead are not correlated with distance from the Libyan shores, our identification strategy provides causal estimates under the standard diff-in-diff assumptions that we outlined in the methodology section of the paper. In what follows we illustrate what are the potential threats to identification that arise from the way the missing migrants data is collected and reported.

Firstly, the probability of observing a deadly event depends on the number of departures from Libya. If more boats depart from Libya, holding everything else constant, the number of people found dead should increase in all the Central Mediterranean. Crucially, as in order to get to the Italian shores, boats need to first survive in areas closer to Libya, in principle, more departures could result in a higher number of deadly events in areas closer to Libya. We address this issue in three different ways: in one of our model specification, we control for an interaction term between the number of departures and quadrant dummies. If the

number of departures is uncorrelated with the timing of the externalization of SAR activities to the LCG, this approach will allow us to fully account for the potential bias induced by a change in departures. Moreover, in [section A.5](#) in the online appendix, we control for a proxy of departures that is less likely to be correlated with the timing of the MoU, the lag of departures and the lag of migration intentions of migrants surveyed in Niger. We also conduct a placebo experiment, which we illustrate in [section A.5](#) in the online appendix, where estimate the effect of a placebo policy in February 2016. Finally, in ?? in the online appendix, we show that the potential bias generated by death incidents in areas far away from the Libyan shore being conditional on survival in areas close to the Libyan shores is two orders of magnitude lower than the estimates of our main specification. Finally, our non-parametric identification strategy clearly shows that the change in the probability of observing a deadly event only occurs in areas close to the Libyan shores.

Secondly, the probability of observing a deadly event depends on the probability of being found after a shipwreck. The probability of being found after a shipwreck depends on the number of non-migrant boats that are in the Mediterranean and are going to report a shipwreck. If there are more Coast Guard or commercial boats in a specific area it is going to be more likely to find migrant boats that shipwrecked. In our setting, we proxy for the number of commercial boats in the Mediterranean using crossings of the Suez Canal, and interacting this measure with quadrant dummies. However, if the likelihood of having an incident reported is correlated with both the timing of the externalization of SAR activities to the LCG and with distance from the Libyan shores, our estimates could be biased. In particular, if the probability of a death incident being reported increases closer to the Libyan shores – relatively to areas that are more distant from the Libyan shores – after the externalization of SAR activities to the LCG, our estimates would be downward biased. On the contrary, if the probability of a deadly incident being reported closer to the Libyan shores decreases, relative to areas that are more distant from the Libyan shores, after the externalization of SAR activities to the LCG, our estimates would be upwards biased – and therefore our estimated negative effect would be a lower bound for the *reduction* in the probability of observing deadly events far away from the Libyan shores –. We argue that the externalization of SAR activities to the LCG, if anything, led to the probability of reporting

a deadly incident being lower for two main reasons. First, since Missing Migrants data rely on media, if anything, the take-up of operations by the newly constituted and under-trained and under-staffed LCG (ENFM 2018), reduced the probability of an incident being reported closer to the Libyan shores. Second, the hostility of the LCG towards NGOs (Al Jazeera 2017) and the decision of some NGOs to cease operations close to the Libyan coast would lower the chances of a death incident being reported closer to the Libyan shores after the externalization of SAR operations to the LCG. We directly test whether our results are driven by increased reporting by Libyan authorities after the externalisation in section A.5, and we find that our results are not affected by it.

Thirdly, the probability of observing a deadly event can be determined by the risk of dying while trying to cross the Mediterranean. We show that after the externalization of SAR activities to the LCG the probability of observing a deadly event in areas that are likely to be patrolled by the LCG increases. In order to interpret our estimates as the result of an increase in the risk of dying while crossing the Mediterranean we need to assume that no other event affected the probability of observing a deadly event close to the Libyan shores after the externalization of SAR activities to the LCG, besides the externalization of SAR activities to the LCG – a standard assumption in any difference-in-differences research design –, along with no correlation between the way that the missing migrants data is collected and reported – which we discussed above –.

A.4 Descriptive Evidence

In this section of the Online Appendix we provide complementary descriptive evidence regarding departures from Libya and overall departures, arrivals and deaths on the Central, Western and Eastern Mediterranean route.

Departures from Libya Departures from Libya follow a seasonal pattern. Departures are generally higher during the summer and lower during the winter. Figure A.1, shows the number of estimated departures from Libya, calculated by the Italian Institute for International Political Studies (ISPI). Departures in 2017 follow a similar pattern to the one observed in 2016, and then drastically drop in July 2017 – after the ONG code of conduct

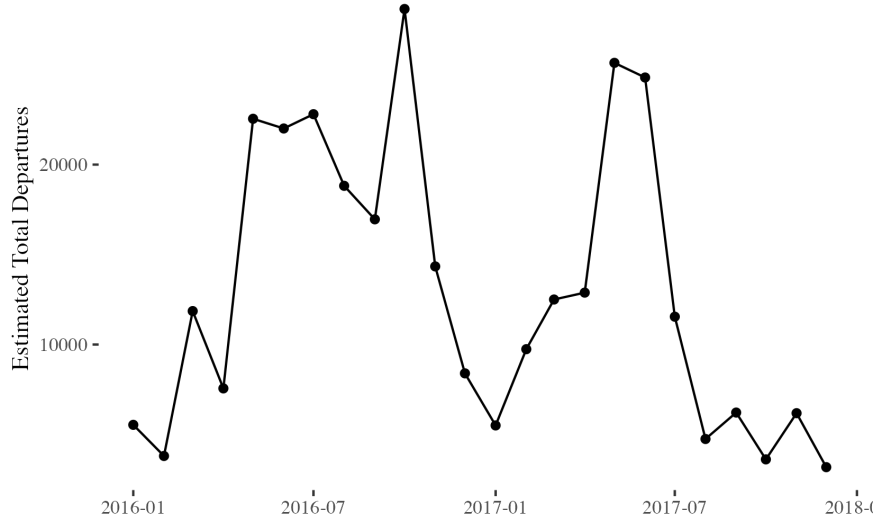


Figure A.1: Number of Estimated Departures from Libya

Notes: The figure shows the number of estimated departures from Libya from 2016 calculated by the Italian Institute for International Political Studies (ISPI)

and the start of training programs by Italian officers in Libya – .

The drop in departures comes immediately after the increase in the probability of observing a deadly event that we document when estimating the dynamic treatment effect of externalizing rescue operations to the Libyan Coastguard.

Figure A.2 shows the difference in departures between 2016 and 2017. Again, it is clear how departures from Libya in 2017 follow a similar pattern to the one observed in 2016, with a drastic drop starting from July 2017.

By comparing these plots with our dynamic treatment effect estimations, estimated according to **Equation 2**, we see that we find the distance from Libyan shores to affect the probability of observing a deadly event during months where departures were similar to the previous year – corroborating the hypothesis that departure behavior did not change immediately after the signing of the MoU –, and we do not find any statistically significant effect after departures drop.

Western, Central, and Eastern Mediterranean Route Arrivals in Europe along the three different Mediterranean routes follow different temporal dynamics. In this section we

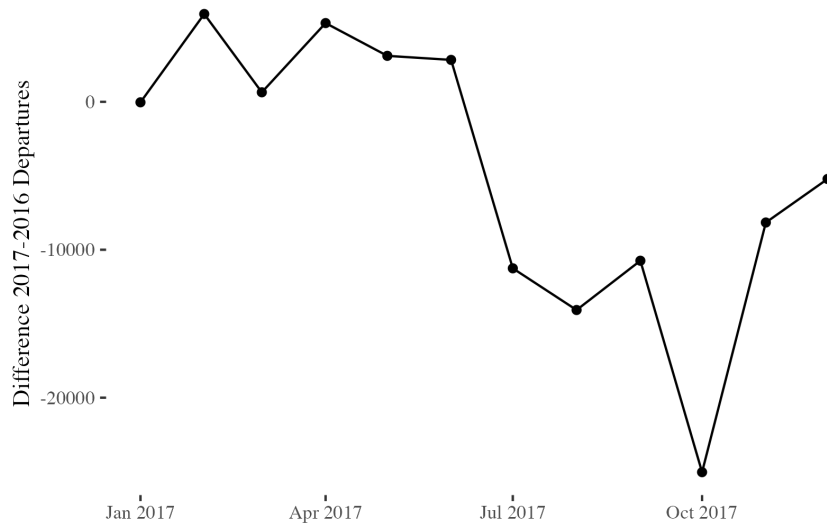


Figure A.2: Number of Estimated Departures from Libya 2017-2016 Difference

Notes: The figure shows the difference in estimated departures from Libya calculated by the Italian Institute for International Political Studies (ISPI) between 2016 and 2017

illustrate the dynamics of arrivals to Europe along the three major routes. As described in the main body of the paper the Western Mediterranean route involves crossing the Mediterranean from Algeria or Morocco to arrive in Spain. The Central Mediterranean route involves crossing the Mediterranean from Libya or Tunisia to arrive to Italy. The Eastern Mediterranean route refers to the migratory route to Greece, Cyprus and Bulgaria, primarily by way of Turkey. ¹²

Figure A.3 shows the number of arrivals in Greece (Eastern Mediterranean Route), Italy (Central Mediterranean Route) and Spain (Western Mediterranean Route). The data starts from when it was firstly made available by the UNHCR and is shown until the end of our sample period. It is important to notice that data for arrivals in Spain starts at a later date than data for Greece and Italy. The three routes differ in the raw number of arrivals, with the Eastern Route having on average 19.190 arrivals per month – as shown in Table A.1 – as the most trafficked route. The number of arrivals displays great variation over time, with arrivals in Greece peaking right at the end of 2015, after the Syrian refugee crisis and before

¹²Our data includes in the Western Mediterranean route also the North-west Africa maritime route, which involves getting to the Canary Islands from Mauritania, Western Sahara, and Morocco

the negotiation of the deal between the EU and Turkey. The number of immigrants arriving to Italy and to Spain are significantly lower, with an average of 10 thousand immigrants arriving to Italy every month and an average of 3000 immigrants arriving to Spain every month – albeit in a different sample period –. Crucially, while from a purely descriptive point of view there is a clear complementarity between arrivals in Italy and arrivals in Spain, arrivals in Italy stay on a similar level when arrivals to Greece drop, showing no evidence of a substitution between the two routes.

[Figure A.3](#) shows the number of monthly arrivals when we sum the arrivals along the three routes. It can be noticed how the pattern is completely dominated by the arrivals to Greece, which were drastically reduced after the EU-Turkey deal.

Arrivals and Deaths on the Western and Central Mediterranean Route We have shown in the paper that the probability of observing a deadly event in areas that were more likely to be patrolled by the Libyan LCG increases after the signing of the MoU. In the second part of the paper we presented evidence in favour of a substitution effect, whereby migrants, after the MoU decide to migrate using the Western Mediterranean Route instead of the Central Mediterranean Route. These circumstances could mean that the total number of deaths – when considering both routes together – could decline, but arrivals could be at a similar level – for example if it is less likely for a deadly event to occur on the Western Mediterranean Route–. In [Figure A.5](#) we report aggregated figures for the total number of arrivals and deaths on both routes, as well as the ratio of the two (i.e. $\frac{N.Deaths}{N.Arrivals}$). We also report average before and after the signing of the MoU (thick black lines). It can be noticed that the total number of deaths – i.e. the sum of the number of deaths for drowning in the Central and Western Mediterranean – stayed relatively stable before and after the signing of the MoU. The average total number of arrivals slightly decreased, and the ratio of total number of deaths over total number of arrivals slightly increased. [Figure A.6](#) shows the same figures when we include the Eastern Mediterranean Route. In this case the numbers are strongly driven by the high number of arrivals to Greece at the end of 2015, but the changes around the signing of the MoU are very similar to those that we can observe when we only consider the Western and Central Mediterranean route.

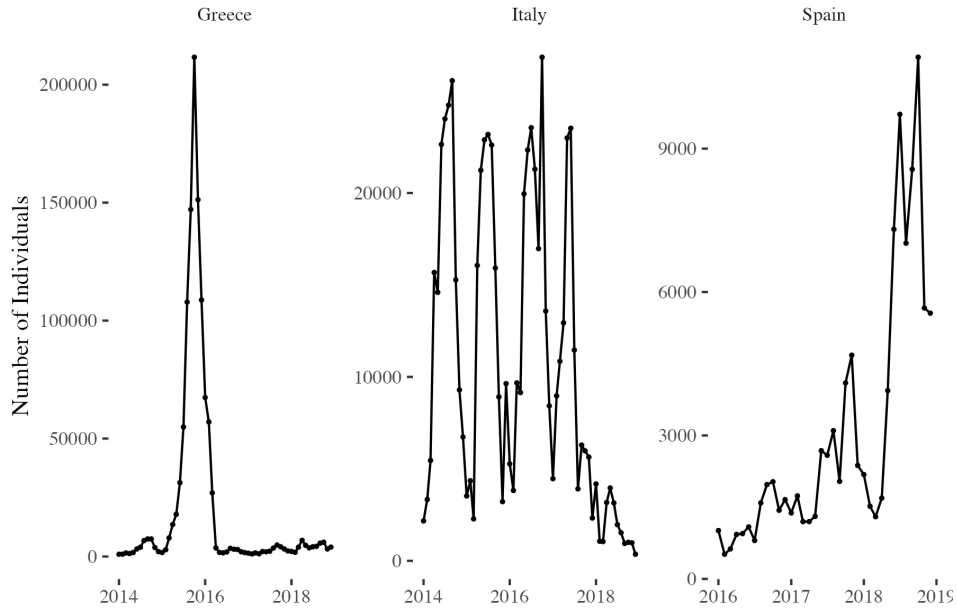


Figure A.3: Number of Arrivals Along Mediterranean Routes

Notes: The figure shows the number of monthly arrivals to Greece (left panel), Italy (central panel) and Spain (right panel) from when data was available from the UNHCR to the end of our sample period.

Table A.1: Descriptive Statistics: Arrivals

Country of Arrival	Mean	Median	SD	Min	Max
Greece	19190.617	3617.0	42060.624	955	211663
Italy	10801.950	8944.0	8456.468	359	27384
Spain	3010.111	1854.5	2702.885	515	10912

Notes: This table shows descriptive statistics for the number of arrivals in Greece Italy and Spain

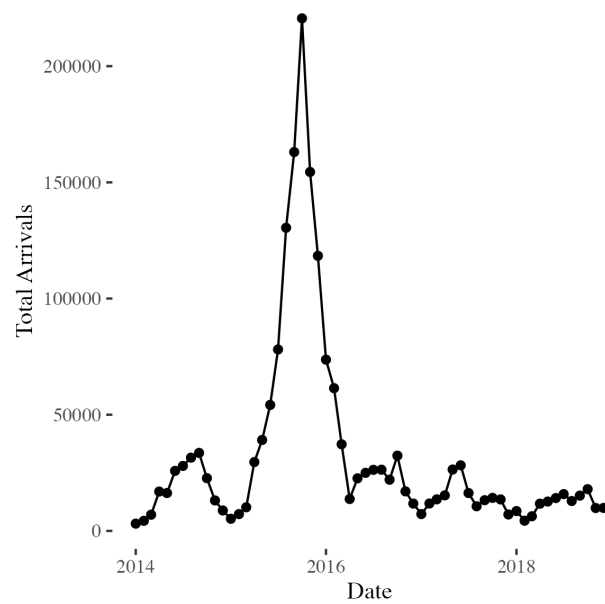


Figure A.4: Sum of Number of Arrivals Along Mediterranean Routes

Notes: The figure shows the sum of monthly arrivals to Greece (left panel), Italy (central panel) and Spain (right panel) from when data was available from the UNHCR to the end of our sample period.

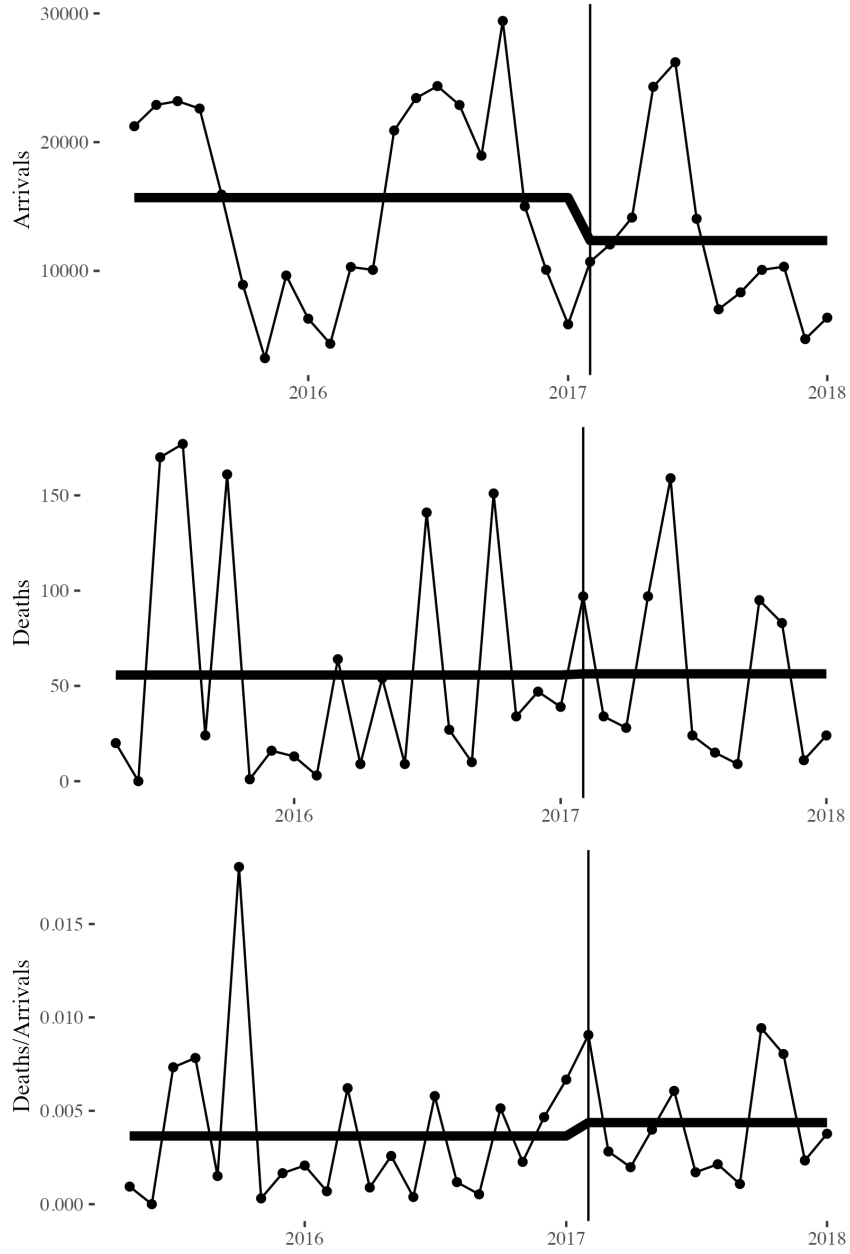


Figure A.5: Total Number of Arrivals and Deaths (Western and Central Mediterranean Route)

Notes: The figure shows the sum of monthly arrivals to Italy and Spain in our sample period (top panel). The total number of deaths on the Central and Western Mediterranean Route (central panel) and the ratio between deaths and arrivals (bottom panel). The thick black lines represent averages before and after the signing of the MoU.

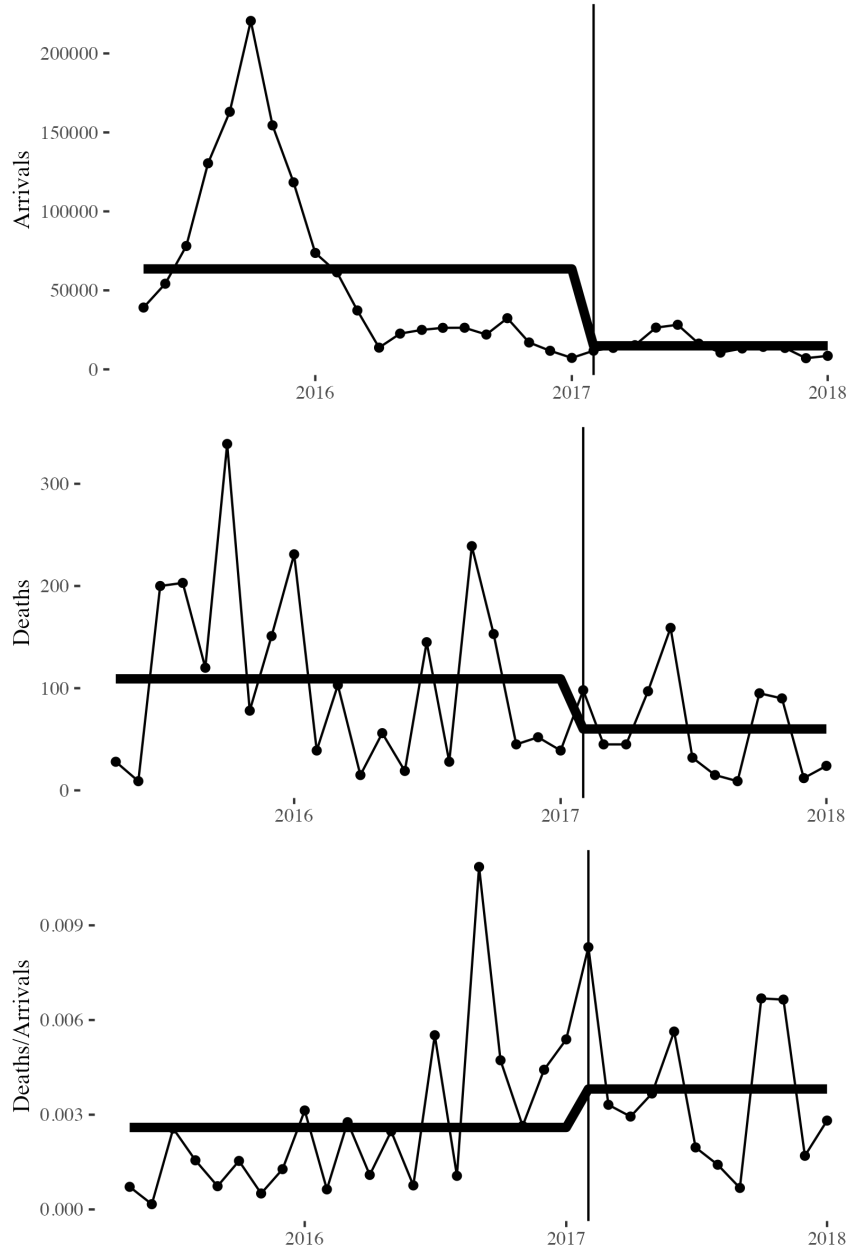


Figure A.6: Total Number of Arrivals and Deaths (Western, Central and Eastern Mediterranean Route)

Notes: The figure shows the sum of monthly arrivals to Italy, Spain and Greece in our sample period (top panel). The total number of deaths on the Central, Western, and Easter Mediterranean Route (central panel) and the ratio between deaths and arrivals (bottom panel). The thick black lines represent averages before and after the signing of the MoU.

A.5 Robustness Checks

Number of Deaths In our main analysis we use a linear probability model to test if the probability of dying in areas that are more likely to be patrolled by the LCG increases. This approach allows us to test if deadly events occur more often in areas that are closer to the Libyan shores, after the externalization of SAR activities to the LCG. However, if migrants start departing in smaller boats after the externalization of SAR activities, the probability of observing a deadly event could increase, while the number of deaths stays constant. Table A.2 shows that our results are qualitatively similar when using the inverse hyperbolic sine (IHS) transformation of the number of deaths as our dependent variable. The IHS transformation is similar to the natural logarithm transformation but allows to account for the prevalence of zeros, while still allowing to interpret estimates as elasticities (Bellemare & Wichman (2020)). The results of this exercise show being far away from the Libyan coast by 100km decreases the number of deaths by 3.6% to 7.9%.

Arbitrary Quadrant Definition Our geographical difference-in-differences analysis relies on our definition of quadrants, which represent the main statistical unit in our regressions. To test how sensitive our estimates are to this choice, we run alternative estimations using quadrants of different measures, respectively of 1, 0.5 and 0.1 degrees of latitude or longitude on each side. Table A.3 presents the results. Quadrants' measure does not influence our results' significance or magnitude relative to the average probability of observing deadly events.

Potential Spillovers and SUTVA violations Another potential issue with our estimates is the fact that the outcome in our treatment units affects the outcome in control units. In fact, the probability of observing deadly events in areas far from Libya likely depends on how many boats wreck closer to Libyan coasts. Ideally, one would rather want to run a difference-in-differences estimation on the *risk of shipwreck* itself; unfortunately, we can not observe it. That said, we are able to model the difference between our estimation strategy and the ideal one, showing that the potential bias due to spillovers is negligible.

In what follows, we compare our estimation strategy — which compares the relative

evolution of the probability of observing shipwrecks closer and farther away from Libya — to an ideal estimation that uses the risk of shipwreck as its main outcome¹³. Our analysis shows that the implicit bias in our analysis is negligible when compared to the magnitude of our estimates.

Consider an environment in which M migrant boats sail from Libya heading for Italy. Once they depart, the boats sail through a first area of the sea — which we call *close* — and are exposed to a risk of shipwreck ρ^{close} . If they get through, they move to a second area *far*, where the risk of wrecking is ρ^{far} .

We assume that the number of boats departing from Libya stays constant over time, so that we can normalise M to 1¹⁴. Therefore, the probability of observing deadly events in each area at time t is

$$\pi_t^{close} = \rho_t^{close} \quad (4)$$

$$\pi_t^{far} = [1 - \rho_t^{close}] \times \rho_t^{far} \quad (5)$$

Comparing the evolution of π^{far} and π^{close} between t and $t+1$ in a difference-in-differences setting amounts to estimating:

$$\tau_\pi = \Delta\pi^{far} - \Delta\pi^{close}, \quad (6)$$

where τ is our difference-in-differences estimator. However, what we would actually like to estimate is

$$\tau_\rho = \Delta\rho^{far} - \Delta\rho^{close}, \quad (7)$$

the relative change in the risk of shipwrecking between the two areas.

As the probability of observing deadly events coincide with the risk of shipwrecking in

¹³We define the risk of shipwreck as the probability that a boat wrecks in a given area of the Mediterranean, *once it actually enters that area*.

¹⁴Allowing attempted crossings to vary over time complicates the analysis, but does not qualitatively change the results.

the *close* area, the bias of our estimator is given by

$$\tau_\pi - \tau_\rho = \Delta\pi^{far} - \Delta\rho^{far}, \quad (8)$$

which simplifies to

$$\tau_\pi - \tau_\rho = \rho_t^{far} - \Delta\rho^{close} \times \rho_t^{far} - \Delta\rho^{far} \times \rho_t^{close}. \quad (9)$$

As we observe both π_t^{close} and π_t^{far} , using Equations 4 and 5 we can compute the change in the product of the risks of shipwrecking in both areas. To do so, we divide our quadrants in the Mediterranean in two areas based on whether their distance from Libya is above or below the median. Moreover, we divide our period of analysis in two around January 2017, when the Memorandum of Understanding was signed.

Using this setting, we compute the risk terms highlighted in the model above. Intuitively, the risks of shipwreck in each area of the Mediterranean is quite small, and even more so are their products¹⁵. The estimated bias term amounts to 0.00013, around two orders of magnitude smaller than our estimates for the impact of the Memorandum of Understanding.

Human Error in Geo-coding deaths We explicitly consider the possibility of human error in geo-coding deadly events in the Central Mediterranean. One might be worried, for example, that when the location of a shipwreck is uncertain our data set might impute a default location of all incidents, if such a process is in place, and varies over-time our estimates would be biased. We address this concern by showing that there is no clear bunching pattern in the reported coordinates of deadly events. We do this by plotting the histograms of longitude and latitude in Figure A.10. This figure shows that the coordinates of deadly events do not show a clear bunching pattern. We can see that deadly events tend to be concentrated in some areas (for example at around 33 degrees of latitude), however, within these areas coordinates are roughly normally distributed. By no surprise, we see a high concentration of deadly events at the latitude at which Libyan shores are located, this feature, however, does not correspond to a huge concentration around any specific longitude.

¹⁵The highest of the risk terms — corresponding to the close area after the signature of the Memorandum of Understanding — equals around 0.06.

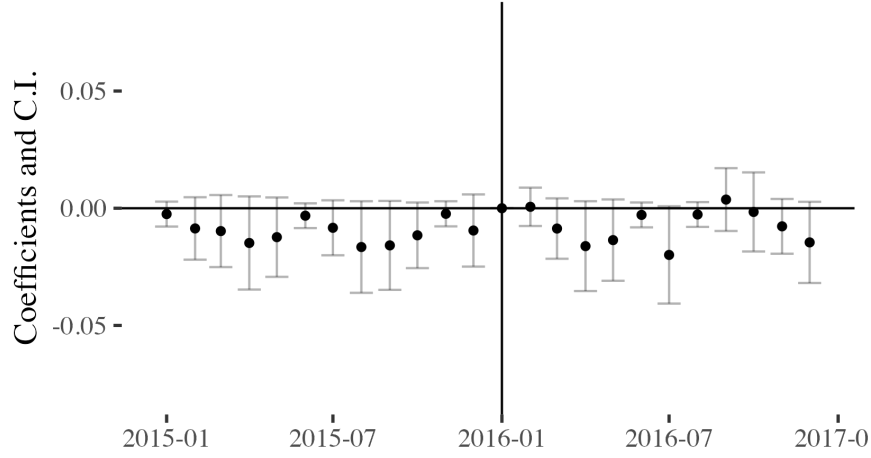


Figure A.7: Flexible difference-in-differences estimation - Placebo Experiment

Notes: The figure displays the results of Equation 2 when estimated on a sample that includes data from 2015 to 2016 and treatment starting in February 2016. The coefficients estimate the effect of the interaction of quadrant's distance from Libya (in hundreds of kilometres) and month dummies, along with the associated 95% confidence intervals.

This pattern is consistent with a high amount of deadly events occurring very close to the Libyan shores.

Placebo Estimate February 2016 As an additional robustness check we estimate dynamic treatment effects on a *placebo* intervention in February 2016. The logic of this approach is that if our estimates would be driven by variation in departures we would estimate a non-zero dynamic treatment effect in months with higher/lower departures during the summer of 2016.

Figure A.7 shows dynamic treatment effect estimates for a placebo intervention that took place in February 2016. The point estimates of dynamic treatment effects are relatively close to 0 and are never statistically significant at the 95% confidence level. This result is consistent with our results not being driven by variation in the number of departures.

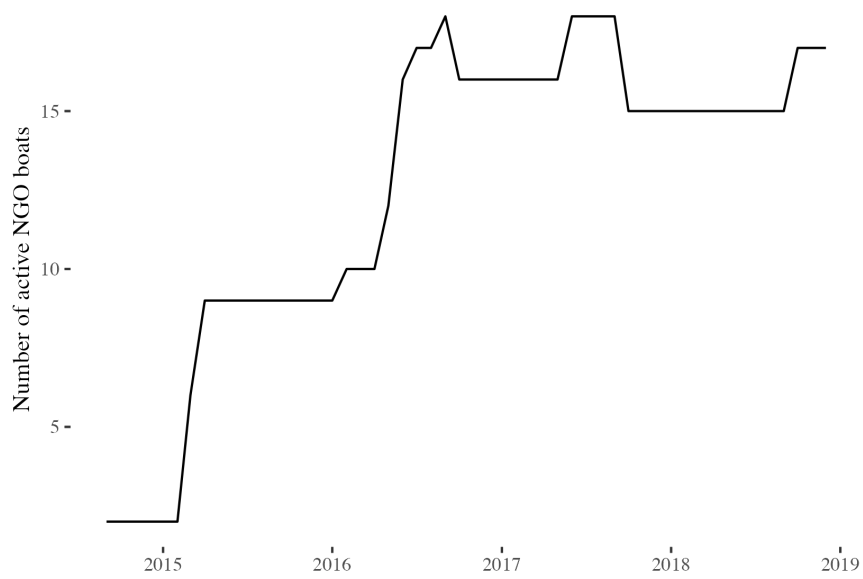


Figure A.8: Number of active NGO boats in the Mediterranean.

Notes: The figure shows the evolution of the number of NGO boats that were active in the Mediterranean, between the end of 2014 and 2020.

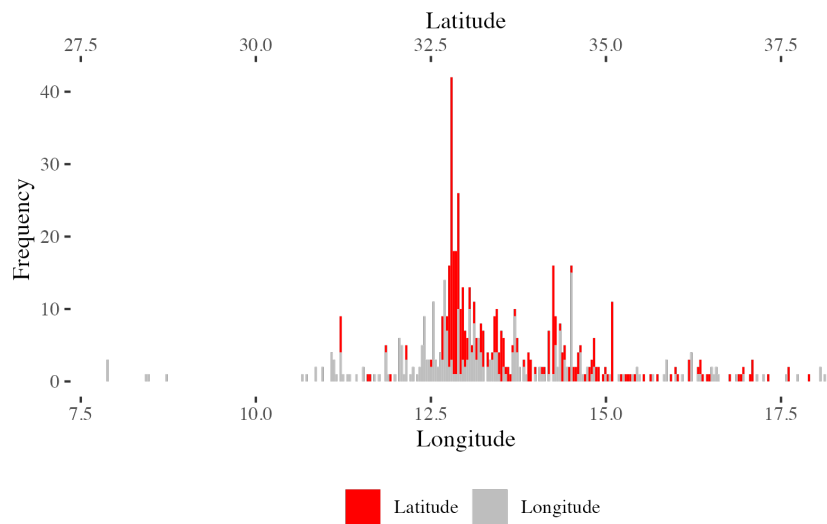


Figure A.10: Histogram of coordinates of deadly events

Notes: This Figure shows a histogram of coordinates for deadly events in the Central Mediterranean.

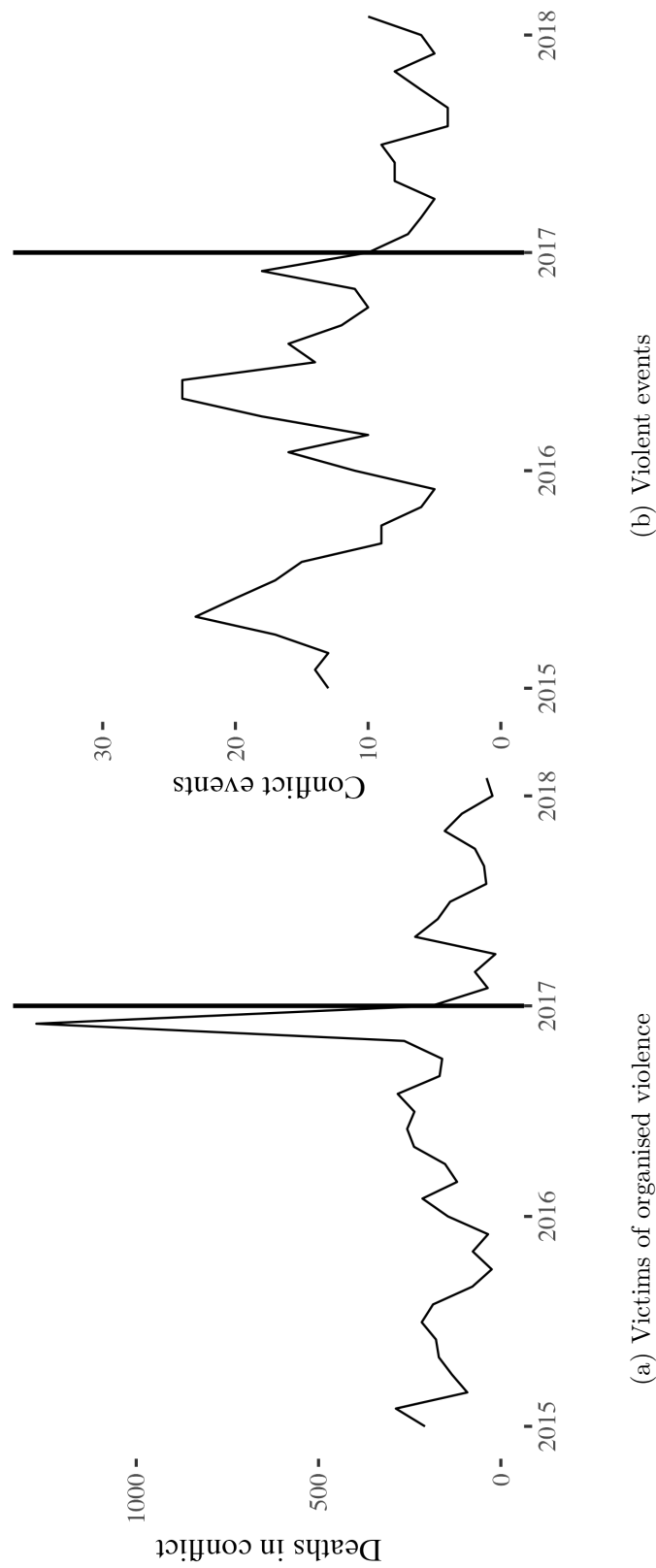


Figure A.9: Conflict in Libya

Notes: The graph displays the evolution of the number of deaths and deadly events related to the Libyan civil war, between 2014 and 2020.

Table A.2: Number of Deaths (in IHS) - Distance

	<i>Dependent variable:</i>						
	Average Number of Monthly Deaths in Central Mediterranean=130.68						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Distance from Libya x Post Deal	-0.036* (0.020)	-0.036* (0.020)	-0.060** (0.027)	-0.062* (0.032)	-0.056** (0.027)	-0.055* (0.028)	-0.079* (0.044)
Log(Conflict events) x Distance			Yes	Yes	Yes		
NGO boats x Distance			Yes	Yes	Yes		
Log(Ships Crossing Suez) x Distance			Yes	Yes	Yes		
Log(Conflict events) x Quadrant FE						Yes	Yes
NGO boats x Quadrant FE						Yes	Yes
Log(Ships Crossing Suez) x Quadrant FE						Yes	Yes
Estimated Number of Departures x Quadrant FE							Yes
Sea conditions			Yes	Yes	Yes	Yes	Yes
Quadrant FE		Yes	Yes	Yes	Yes	Yes	Yes
Date FE		Yes	Yes	Yes	Yes	Yes	Yes
Quadrant time trends				Yes			
Distance-weighted number of deaths x Date dummies					Yes	Yes	Yes
Distance-weighted sea conditions x Date dummies					Yes	Yes	Yes
Fraction of Inflatables X Distance			Yes	Yes	Yes	Yes	Yes
Observations	3,382	3,382	3,382	3,382	3,382	3,382	2,136
Adjusted R ²	0.018	0.080	0.081	0.116	0.094	0.124	0.154

The table shows the outcome of several difference-in-differences regressions in which the outcome is the number of deaths in quadrant q in month t . The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Table A.3: Robustness check - Quadrants' dimension

	<i>Dependent variable:</i>		
	q = 1x1 (1)	q = 0.5x0.5 (2)	q = 0.1x0.1 (3)
Distance from Libya x Post Deal	-0.010** (0.005)	-0.005** (0.002)	-0.0002*** (0.0001)
Mean Prob. of monthly events	0.02	0.01	4e-04
Observations	3,382	10,564	207,252
Adjusted R ²	0.022	0.009	0.0004

The table shows the outcome of several difference-in-differences regressions in which the outcome is a dummy for whether at least one deadly event is observed in quadrant q in month t . The regressor of interest is the interaction between the quadrant's log distance from Libya and a dummy for months after January 2017. In each column, we divide the Central Mediterranean sea in quadrants of different dimensions: 1,0.5 and 0.1 degrees of latitude times 1, 0.5 and 0.1 degrees of longitude, respectively. Standard errors are clustered by quadrant. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Alternative Control for Departures In our main specification we add departures from Libya to control for systematic changes in the probability of observing a deadly event that are correlated both with the distance from Libyan shores and with the number of departures. Our main specification already includes date fixed effects, which allow us to only exploit spatial variation in the location of deadly events. Therefore, we control for an interaction between quadrant fixed effects and number of departures, to allow the effect of departures to vary by quadrant. This approach, however, raises the issue of introducing a bad control, as departures from Libya can also be influenced by the externalization of SAR operations to the LCG.

We address this issue in three different ways. Firstly, we control for the lag of log of the number of departures, interacted with quadrant dummies. Secondly, we control for the log of the number of individuals surveyed in Niger, who declared that they wanted to migrate to Libya, in the quarter prior to the quarter where we observe a specific deadly event – the lag of individuals who were about to arrive to Libya –. We use this variable as a proxy for departures, which is plausibly less affected by the externalization of SAR

activities to the LCG than the number of departures from Libya. Finally, we estimate a model specification where we divide our outcome variable by the total number of departures. This approach allow us to assign a different weight to deadly events, weighting deadly events more when departures are low, compared to deadly events when departures are high. This modeling choice explicitly addresses the consideration that departures could be affected by the externalization of SAR activities to the LCG by including departures in the dependent variable.

[Table A.4](#) shows the results of adding the interaction term between the lag of departures and quadrant dummies. In particular, we show estimates for a one month, two months and three months lag in columns (1), (2) and (3) using our main specification – column (6) in [Table 1](#) –. The results remain largely unchanged, mitigating the concerns of a *bad control* bias, as we would expect the bias to be different for different lags of total departures.

[Table A.5](#) shows the results of adding the interaction term between the log of the number of individuals surveyed in Niger, who declared that they wanted to migrate to Libya. In particular, we show estimates for a one quarter, two quarters and three quarters lag in columns (1), (2) and (3) using our main specification – column (6) in [Table 1](#) –. The results remain largely unchanged, suggesting that the change in the risk of dying in areas closer to the Libyan shores is driving our results.

[Table A.6](#) and [Table A.7](#) show the results of dividing our outcome of interest by the total number of estimated departures. In particular, [Table A.6](#) shows the results for several specifications of our main outcome variable and [Table A.7](#) shows the same result when we substitute our dummy variable for the presence of a deadly event in a quadrant with the inverse hyperbolic sine transformation of the number of deaths. Overall, the results remain statistically significant and similar in relative magnitude to our main estimates. The only exception is the results that we obtained when adding quadrant specific time trends (column 4) of tables [Table A.6](#) and [Table A.7](#). It is important to note that with the inclusion of quadrant-specific time trends, the standard errors of the estimated coefficients double compared to our baseline estimates. This likely reflects the impact of introducing a linear trend to a highly nonlinear variable such as the occurrence of a fatal event in a specific quadrant.

Effect of Externalization on Reporting The decision to externalize rescue operations to the LCG likely reduced the presence of boats near the Libyan shores, which initially raised concerns about potential underreporting of deaths in the area. However, it is possible that this externalization led to an increased presence of vessels capable of reporting incidents near the Libyan shores. If the latter scenario holds true, our reported estimates in [Table 1](#) may be skewed, overestimating the reduction in the likelihood of observing deadly events near the Libyan shores. To address this concern, we recalculated our estimates using only incidents reported by sources with no direct link to the LCG. While this adjustment may underestimate the actual reduction in risk due to externalization, by excluding incidents inherently closer to the Libyan coast, it helps to rule out the possibility that our findings are solely influenced by changes in reporting by the Libyan authorities. Specifically, we removed 191 incidents from our analysis, including all reports from IOM Libya (118 incidents), as well as those from other organizations based in Libya, such as the Libyan Red Crescent (4 incidents) and the Libyan Coast Guard (2 incidents).

The results of this adjusted analysis are presented in [Table A.8](#). Notably, these findings closely resemble those in [Table 1](#), albeit with slightly reduced magnitudes. It's important to acknowledge that the estimates in [Table A.8](#) could be mechanically smaller due to the removal of deadly events inherently closer to the Libyan shores.

Table A.4: Probability of monthly deaths - Distance: Controlling for Lag of Departures

	<i>Dependent variable:</i>		
	Prob. of monthly deaths (Mean=0.02)		
	(1)	(2)	(3)
Distance x Post Deal	-0.024** (0.009)	-0.025** (0.011)	-0.021** (0.010)
Log(Lag Departures) x Quadrant FE	Lag 1	Lag 2	Lag3
Lag Departures x Quadrant FE	Yes	Yes	Yes
NGO boats x Quadrant FE	Yes	Yes	Yes
Log(Ships Crossing Suez) x Quadrant FE	Yes	Yes	Yes
Sea conditions	Yes	Yes	Yes
Quadrant FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Distance-weighted number of deaths x Date dummies	Yes	Yes	Yes
Distance-weighted sea conditions x Date dummies	Yes	Yes	Yes
Observations	2,047	1,958	1,869
Adjusted R ²	0.149	0.163	0.148

The table shows the outcome of several difference-in-differences regressions in which the outcome is a dummy for whether we observe any deaths (or missing persons) in quadrant q in month t . The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. In all specifications we control for an interaction term between the lag of the logarithmic transformation of the total number of departures and quadrant dummies. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Table A.5: Probability of monthly deaths - Distance: Controlling for Migrant's Decisions in Niger

	<i>Dependent variable:</i>		
	Prob. of monthly deaths (Mean=0.02)		
	(1)	(2)	(3)
Distance x Post Deal	-0.025** (0.010)	-0.028*** (0.011)	-0.020* (0.010)
Log(Quarterly Leave To Libya) x Quadrant FE	Lag 1	Lag 2	Lag3
Log(Conflict events) x Quadrant FE	Yes	Yes	Yes
NGO boats x Quadrant FE	Yes	Yes	Yes
Log(Ships Crossing Suez) x Quadrant FE	Yes	Yes	Yes
Sea conditions	Yes	Yes	Yes
Quadrant FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Distance-weighted number of deaths x Date dummies	Yes	Yes	Yes
Distance-weighted sea conditions x Date dummies	Yes	Yes	Yes
Observations	2,047	1,780	1,513
Adjusted R ²	0.182	0.201	0.288

The table shows the outcome of several difference-in-differences regressions in which the outcome is a dummy for whether we observe any deaths (or missing persons) in quadrant q in month t . The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. In all specifications we control for an interaction term between the quarterly lag of the logarithmic transformation of the number of migrants that are in Niger and declare that they want to migrate to Libya and quadrant dummies. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Table A.6: Probability of monthly deaths scaled by number of departures

	<i>Dependent variable:</i>					
	(Any Death in Quadrant/Departures)*100,000 (Mean=0.23)					
	(1)	(2)	(3)	(4)	(5)	(6)
Distance x Post Deal	-0.139*** (0.054)	-0.139** (0.054)	-0.147** (0.068)	-0.051 (0.113)	-0.159** (0.072)	-0.150** (0.072)
Log(Conflict events) x Distance			Yes	Yes	Yes	
NGO boats x Distance			Yes	Yes	Yes	
Log(Ships Crossing Suez) x Distance			Yes	Yes	Yes	
Log(Conflict events) x Quadrant FE						Yes
NGO boats x Quadrant FE						Yes
Log(Ships Crossing Suez) x Quadrant FE						Yes
Estimated Number of Departures x Quadrant FE						
Sea conditions			Yes	Yes	Yes	Yes
Quadrant FE		Yes	Yes	Yes	Yes	Yes
Date FE		Yes	Yes	Yes	Yes	Yes
Quadrant time trends				Yes		
Distance-weighted number of deaths x Date dummies					Yes	Yes
Distance-weighted sea conditions x Date dummies					Yes	Yes
Fraction of Inflatables X Distance			Yes	Yes	Yes	Yes
Observations	2,136	2,136	2,136	2,136	2,136	2,136
Adjusted R ²	0.018	0.067	0.066	0.099	0.138	0.152

The table shows the outcome of several difference-in-differences regressions in which the outcome is a dummy for whether we observe any deaths (or missing persons) in quadrant q in month t , divided by the total number of estimated departures from Libya, and then multiplied by 100,000. The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Table A.7: Number of monthly deaths scaled by number of departures

	<i>Dependent variable:</i>					
	(IHS number of deaths/Departures)*100,000 (Mean=0.84)					
	(1)	(2)	(3)	(4)	(5)	(6)
Distance x Post Deal	-0.504** (0.233)	-0.504** (0.233)	-0.595* (0.318)	-0.276 (0.505)	-0.640* (0.358)	-0.599* (0.360)
Log(Conflict events) x Distance			Yes	Yes	Yes	
NGO boats x Distance			Yes	Yes	Yes	
Log(Ships Crossing Suez) x Distance			Yes	Yes	Yes	
Log(Conflict events) x Quadrant FE						Yes
NGO boats x Quadrant FE						Yes
Log(Ships Crossing Suez) x Quadrant FE						Yes
Estimated Number of Departures x Quadrant FE						
Sea conditions			Yes	Yes	Yes	Yes
Quadrant FE		Yes	Yes	Yes	Yes	Yes
Date FE		Yes	Yes	Yes	Yes	Yes
Quadrant time trends				Yes		
Distance-weighted number of deaths x Date dummies					Yes	Yes
Distance-weighted sea conditions x Date dummies					Yes	Yes
Fraction of Inflatables X Distance			Yes	Yes	Yes	Yes
Observations	2,136	2,136	2,136	2,136	2,136	2,136
Adjusted R ²	0.017	0.061	0.060	0.089	0.068	0.076

The table shows the outcome of several difference-in-differences regressions in which the outcome is the inverse hyperbolic sine transformation of the number of deaths in quadrant q in month t , divided by the total number of estimated departures from Libya, and then multiplied by 100,000.. The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.

Table A.8: Monthly probability of observing deaths - No Libyan Sources

	<i>Dependent variable:</i>						
	Prob. of monthly deaths (Mean=0.02)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Distance x Post Deal	-0.007** (0.004)	-0.007* (0.004)	-0.012** (0.006)	-0.017** (0.007)	-0.010* (0.006)	-0.010* (0.006)	-0.022** (0.009)
Log(Conflict events) x Distance			Yes	Yes	Yes		
NGO boats x Distance			Yes	Yes	Yes		
Log(Ships Crossing Suez) x Distance			Yes	Yes	Yes		
Log(Conflict events) x Quadrant FE						Yes	Yes
NGO boats x Quadrant FE						Yes	Yes
Log(Ships Crossing Suez) x Quadrant FE						Yes	Yes
Estimated Number of Departures x Quadrant FE							Yes
Sea conditions			Yes	Yes	Yes	Yes	Yes
Quadrant FE		Yes	Yes	Yes	Yes	Yes	Yes
Date FE		Yes	Yes	Yes	Yes	Yes	Yes
Quadrant time trends				Yes			
Distance-weighted number of deaths x Date dummies					Yes	Yes	Yes
Distance-weighted sea conditions x Date dummies					Yes	Yes	Yes
Fraction of Inflatables X Distance			Yes	Yes	Yes	Yes	Yes
Observations	3,382	3,382	3,382	3,382	3,382	3,382	2,136
Adjusted R ²	0.016	0.083	0.086	0.099	0.129	0.133	0.187

The table shows the outcome of several difference-in-differences regressions in which the outcome is a dummy for whether we observe any deaths (or missing persons) in quadrant q in month t . The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. In this sample we exclude all deadly events that were reported by Libyan sources. Standard errors are clustered by quadrant. *p-values*: * < 0.1; ** < 0.05; *** < 0.01.